Course notes

Sums and transformations

For a finite random variable X, taking the values x_i with probabilities p_i,

$$E(X) = \sum_i x_i p_i, \qquad V(X) = \sum_i (x_i - E(X))^2 p_i = E(X^2) - E(X)^2.$$

The standard deviation is $\sigma(X) = \sqrt{V(X)}$. Expanding the square in $V(aX+b)$, the shift b cancels and the factor a is squared. For finite random variables X, Y and real numbers a, b,

$$E(X+Y) = E(X) + E(Y), \quad E(aX+b) = aE(X) + b, \quad V(aX+b) = a^2 V(X).$$

The linearity of the expected value does not require independence. If X, Y are independent, $V(X+Y) = V(X) + V(Y)$ (accepted without proof). Without this assumption, this formula cannot be applied. The standard deviation of $aX+b$ is $|a|\sigma(X)$.

Example of dependence. If $Y = X$, then $V(X+Y) = V(2X) = 4V(X)$, and not $2V(X)$ as soon as $V(X) \neq 0$.

Proof for the binomial distribution. Write $X = B_1 + \cdots + B_n$ with the independent indicators of the successes. Each B_i satisfies $E(B_i) = p$, $E(B_i^2) = p$ and $V(B_i) = p - p^2$. By linearity and additivity under independence, $E(X) = np$, $V(X) = np(1-p)$.

Sample and mean

A sample of size n of a distribution is a family of n independent random variables with the same distribution. If they have expected value μ and variance σ^2, we write $S_n = \sum_{i=1}^n X_i$ and $M_n = \frac{S_n}{n}$.

> **Key point · Sum and mean.**
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> $$E(S_n) = n\mu, \quad V(S_n) = n\sigma^2, \qquad E(M_n) = \mu, \quad V(M_n) = \frac{\sigma^2}{n}, \quad \sigma(M_n) = \frac{\sigma}{\sqrt{n}}.$$

To divide the standard deviation of the mean by 2, multiply the size by 4. The variance of the sum increases; that of the mean decreases.

Concentration

Key point · Bienaymé–Chebyshev inequality. For a random variable with expected value μ, variance σ^2, and $\delta > 0$,

$$P(|X - \mu| \geq \delta) \leq \frac{\sigma^2}{\delta^2}.$$

An upper bound greater than 1 remains true, but gives no further information; it can be replaced by 1.

Proof in the finite case. In $V(X) = \sum_x (x - \mu)^2 P(X = x)$, keep only the values satisfying $|x - \mu| \geq \delta$. Each square is then at least δ^2. Hence $V(X) \geq \delta^2 P(|X - \mu| \geq \delta)$.

Law of large numbers

Applied to the mean of a sample,

$$P(|M_n - \mu| \geq \delta) \leq \frac{\sigma^2}{n\delta^2} \xrightarrow[n\to\infty]{} 0.$$

This is the law of large numbers in the setting considered: for a fixed precision, the probability of a deviation at least equal to this precision tends to zero. This result does not say that each realisation approaches μ monotonically.

Method · Choosing a sufficient size. To guarantee $P(|M_n - \mu| < \delta) \geq 1 - \alpha$, it suffices to choose

$$n \geq \frac{\sigma^2}{\alpha\delta^2}.$$

Take the ceiling. This gives the smallest integer satisfying *this bound*, not necessarily the minimum size for the exact probability.

For a frequency of successes, $\mu = p$ and $\sigma^2 = p(1 - p) \leq \frac{1}{4}$. If p is unknown, $n \geq \frac{1}{4\alpha\delta^2}$ is a uniform guarantee.